Sumaiya Afreen

Data Science — Machine Learning — AI Applications

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Profile Summary

Third-year Computer Science student (CGPA: 9.1) with hands-on experience in Machine Learning and AI application development. Built and deployed ML systems using Python, Scikit-learn, TensorFlow, FastAPI, and Streamlit, with experience in model evaluation, RAG pipelines, and data analysis.

Education

B. S. Abdur Rahman Crescent Institute of Science and Technology **2024 – 2028**
B.Tech in Computer Science and Engineering — CGPA: 9.1/10 upto semester 4 *Chennai, India*

Indian Institute of Technology Madras **2024 – Present**
B.S. in Data Science and Applications (Online) *Chennai, India*

Experience

Machine Learning Intern – FlyRank.ai *June 2026 – September 2026*

- Built a **Content Opportunity Scoring system** using search, traffic, engagement, and freshness signals; engineered and audited features from a **large-scale content performance warehouse**.
- Trained an interpretable **Logistic Regression ranking model** that outperformed the rule-based baseline, achieving **1.00 Precision@50** and **0.898 Average Precision**, with grouped validation and leakage checks.

Projects

Car Price Prediction – Used Vehicle Price Modeling — *Python, Scikit-learn, XGBoost* GitHub

- Built an end-to-end regression pipeline on **10,000+ used-car records**; XGBoost achieved an R^2 **score of 0.9406**, outperforming Random Forest (0.9339) and Linear Regression (0.8918).

Handwritten Digit Recognition – CNN Classification — *Python, TensorFlow/Keras, NumPy, Streamlit* GitHub

- Trained a **CNN** on the MNIST dataset, achieving **99%+ test accuracy**, and deployed the model as a Streamlit application with multi-digit recognition and confidence scores.

Research Pilot – AI Research Assistant *(Ongoing)* — *Python, FastAPI, React.js, PostgreSQL, LangChain* GitHub

- Developing an AI-powered research assistant integrating **5+ academic APIs** including OpenAlex, Semantic Scholar, arXiv, Crossref, and PubMed for automated paper discovery and semantic search.
- Building a **dual-agent RAG pipeline** with AI-based fact verification, **source-backed citations**, and an interactive knowledge graph for transparent research exploration.

Technical Skills

Programming Languages: Python, SQL, Java, C++

Data Analysis & Visualization: Microsoft Excel, Power BI, Tableau, Pandas, NumPy

Machine Learning: Scikit-learn, Regression, Classification, Feature Engineering, EDA

Web: React.js, HTML/CSS

Leadership & Activities

- **Joint Secretary - Crescent Readers Club** *2025 – Present*
  Organize reading sessions and member engagement for the club.
- **Design Head - GIR WebOps Team (IIT Madras)** *2024 – 2025*
  Designed promotional assets and branding for technical events.

Certifications & Technical Engagement

- **Oracle Cloud Infrastructure 2025 AI Foundations Associate** – Oracle
- **Machine Learning with Python** – IBM
- **Elite Her Hackathon – Top 250 Finalist** – Elite Coders